

islr-ch5

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```
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```

Chapter V - Resampling Methods

All the questions are as per the ISL seventh printing of the First edition^[^fn:1].

Common

Instead of using the standard functions, we will leverage the `mlr3` package^[^fn:2].

```
#install.packages("mlr3","data.table","mlr3viz","mlr3learners")
```

Actually for `R` version `3.6.2`, the steps to get it working were a bit more involved.

```
install.packages("remotes","data.table",
                 "GGally","precerec") # For plots
```

```
library(remotes)
remotes::install_github("mlr-org/mlr3")
remotes::install_github("mlr-org/mlr3viz")
remotes::install_github("mlr-org/mlr3learners")
```

Load `ISLR` and other libraries.

```
libsUsed<-c("dplyr","ggplot2","tidyverse",
            "ISLR","caret","MASS",
            "pROC","mlr3","data.table",
            "mlr3viz","mlr3learners")
invisible(lapply(libsUsed, library, character.only = TRUE))
```

Question 5.5 - Page 198

In Chapter 4, we used logistic regression to predict the probability of `default` using `income` and `balance` on the `Default` data set. We will now estimate the test error of this logistic regression model using the validation set approach. Do not forget to set a random seed before beginning your analysis.

(a) Fit a logistic regression model that uses `income` and `balance` to predict `default`.

(b) Using the validation set approach, estimate the test error of this model. In order to do this, you must perform the following steps:

1. Split the sample set into a training set and a validation set.
2. Fit a multiple logistic regression model using only the training observations.

3. Obtain a prediction of default status for each individual in the validation set by computing the posterior probability of default for that individual, and classifying the individual to the `default` category if the posterior probability is greater than **0.5**.
 4. Compute the validation set error, which is the fraction of the observations in the validation set that are misclassified.
- (c)** Repeat the process in (b) three times, using three different splits of the observations into a training set and a validation set. Comment on the results obtained.
- (d)** Now consider a logistic regression model that predicts the probability of `default` using `income`, `balance`, and a dummy variable for `student`. Estimate the test error for this model using the validation set approach. Comment on whether or not including a dummy variable for `student` leads to a reduction in the test error rate.

Answer

We will need our data.

```
defDat<-ISLR::Default
```

- Very quick peek

```
defDat %>% summary
```

```
## default      student      balance      income
## No :9667      No :7056      Min.   :  0.0      Min.   : 772
## Yes: 333      Yes:2944      1st Qu.: 481.7      1st Qu.:21340
##                               Median : 823.6      Median :34553
##                               Mean   : 835.4      Mean   :33517
##                               3rd Qu.:1166.3      3rd Qu.:43808
##                               Max.   :2654.3      Max.   :73554
```

```
defDat %>% str
```

```
## 'data.frame':    10000 obs. of  4 variables:
## $ default: Factor w/ 2 levels "No","Yes": 1 1 1 1 1 1 1 1 1 ...
## $ student: Factor w/ 2 levels "No","Yes": 1 2 1 1 1 2 1 2 1 1 ...
## $ balance: num  730 817 1074 529 786 ...
## $ income : num  44362 12106 31767 35704 38463 ...
```

a) Logistic Model with mlr3

Following the [new approach](#) which leverages R6 features leads us to define a classification task first. As far as I can tell, the data needs to be filtered to contain only the things we need to predict with, in this case we are required to use only income and balance so we will do so.

```
set.seed(1984)
redDat<-defDat %>% subset(select=c(income,balance,default))
tskLogiFull=TaskClassif$new(id="credit",backend=redDat,target="default")
print(tskLogiFull)
```

```
## <TaskClassif:credit> (10000 x 3)
## * Target: default
## * Properties: twoclass
## * Features (2):
##   - dbl (2): balance, income
```

This can be visualized neatly as well.

```
autoplot(tskLogiFull)
```

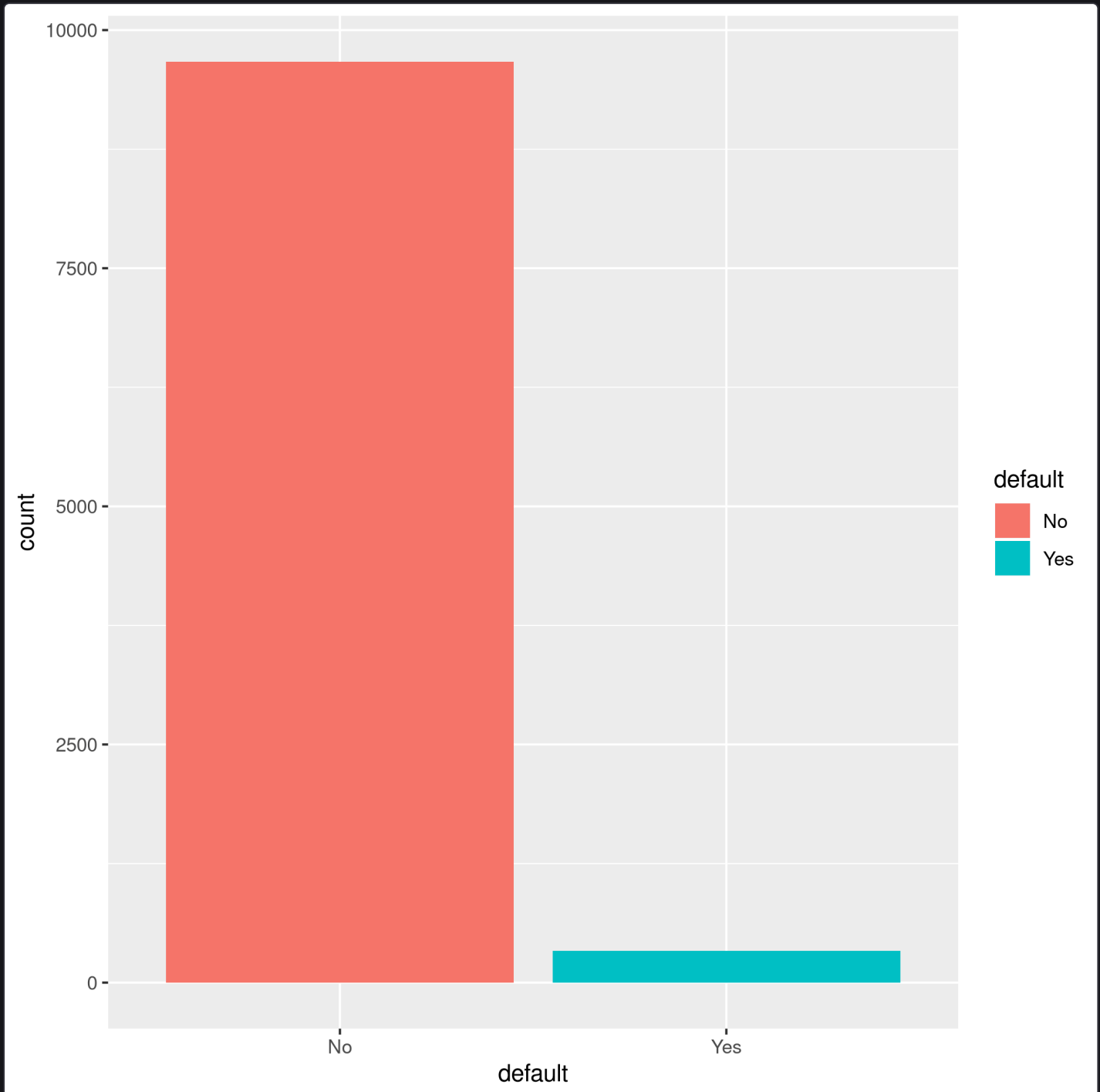


Figure 1: MLR3 Visualizations

We have a pretty imbalanced data-set.

```
autoplot(tskLogiFull,type="pairs")
```

```
## Registered S3 method overwritten by 'GGally':
##   method from
##   +.gg      ggplot2
```

```
## `stat_bin()` using `bins = 30`. Pick better value with `binwidth`.
## `stat_bin()` using `bins = 30`. Pick better value with `binwidth`.
```

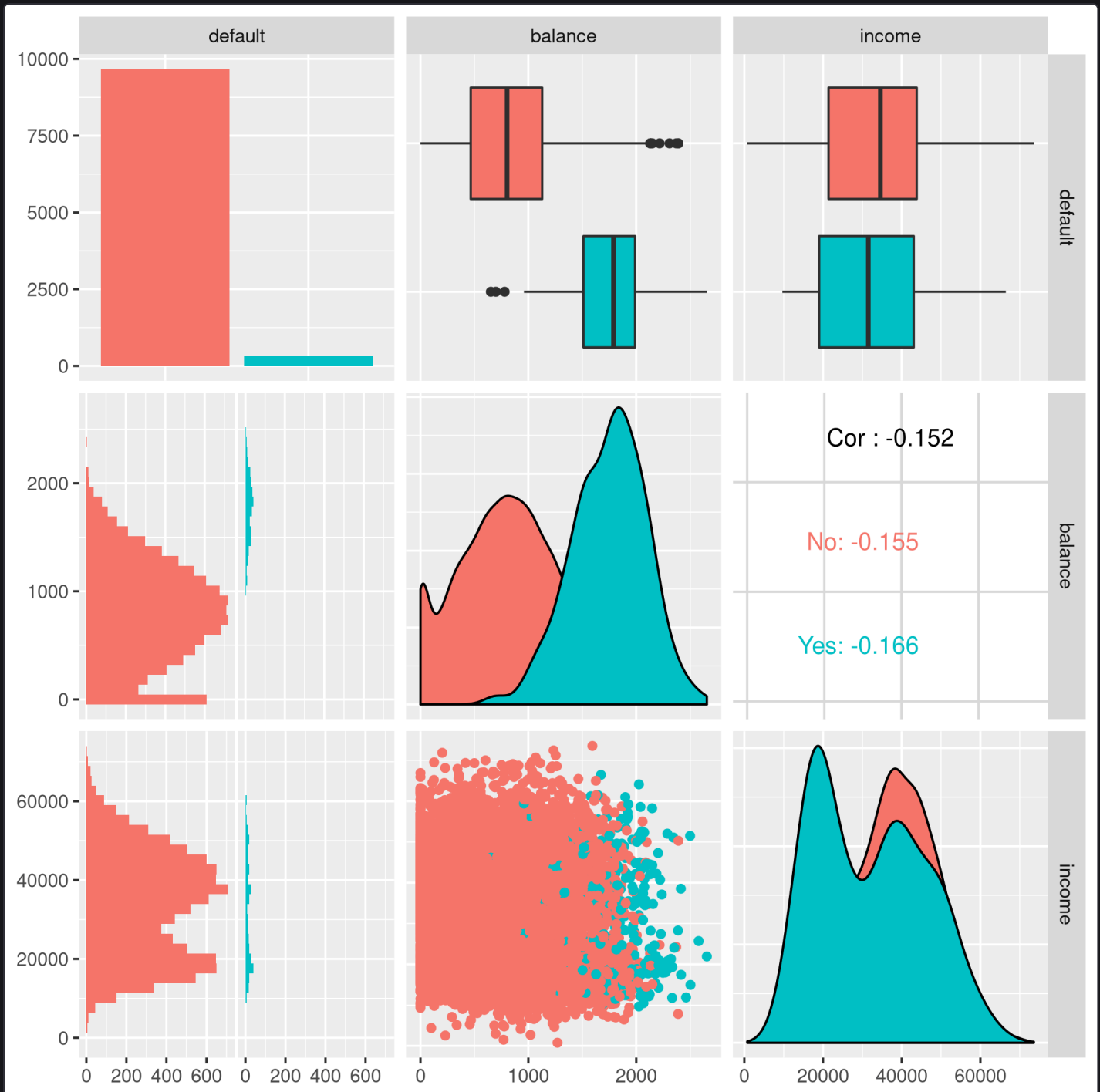


Figure 2: Paired mlr3 data

We can use any of the learners implemented, so it is a good idea to take a quick peek at them all.

```
as.data.table(mlr_learners)
```

```

##          key          feature_types
## 1:      classif.debug logical,integer,numeric,character,factor,ordered
## 2: classif.featureless logical,integer,numeric,character,factor,ordered
## 3:      classif.glmnet          logical,integer,numeric
## 4:      classif.kknn          logical,integer,numeric,factor,ordered
## 5:      classif.lda          logical,integer,numeric,factor,ordered
## 6:      classif.log_reg logical,integer,numeric,character,factor,ordered
## 7: classif.naive_bayes          logical,integer,numeric,factor
## 8:      classif.qda          logical,integer,numeric,factor,ordered
## 9:      classif.ranger logical,integer,numeric,character,factor,ordered
## 10:      classif.rpart          logical,integer,numeric,factor,ordered
## 11:      classif.svm          logical,integer,numeric
## 12:      classif.xgboost          logical,integer,numeric
## 13:      regr.featureless logical,integer,numeric,character,factor,ordered
## 14:      regr.glmnet          logical,integer,numeric
## 15:      regr.kknn          logical,integer,numeric,factor,ordered
## 16:      regr.km          logical,integer,numeric
## 17:      regr.lm          logical,integer,numeric,factor
## 18:      regr.ranger logical,integer,numeric,character,factor,ordered
## 19:      regr.rpart          logical,integer,numeric,factor,ordered
## 20:      regr.svm          logical,integer,numeric
## 21:      regr.xgboost          logical,integer,numeric
##          key          feature_types
##          packages
## 1:
## 2:
## 3:      glmnet
## 4:      kknn
## 5:      MASS
## 6:      stats
## 7:      e1071
## 8:      MASS
## 9:      ranger
## 10:      rpart
## 11:      e1071
## 12:      xgboost
## 13:      stats
## 14:      glmnet
## 15:      kknn
## 16: DiceKriging
## 17:      stats
## 18:      ranger
## 19:      rpart
## 20:      e1071
## 21:      xgboost
##          packages
##          properties
## 1:          missings,multiclass,twoclass
## 2:      importance,missings,multiclass,selected_features,twoclass
## 3:          multiclass,twoclass,weights
## 4:          multiclass,twoclass
## 5:          multiclass,twoclass,weights
## 6:          twoclass,weights
## 7:          multiclass,twoclass
## 8:          multiclass,twoclass,weights
## 9:      importance,multiclass,oob_error,twoclass,weights
## 10: importance,missings,multiclass,selected_features,twoclass,weights
## 11:          multiclass,twoclass
## 12:      importance,missings,multiclass,twoclass,weights
## 13:      importance,missings,selected_features
## 14:          weights
## 15:
## 16:
## 17:          weights
## 18:      importance,oob_error,weights

```

```
## 19: importance,missings,selected_features,weights
## 20:
## 21: importance,missings,weights
##
##      properties
## predict_types
## 1: response,prob
## 2: response,prob
## 3: response,prob
## 4: response,prob
## 5: response,prob
## 6: response,prob
## 7: response,prob
## 8: response,prob
## 9: response,prob
## 10: response,prob
## 11: response,prob
## 12: response,prob
## 13: response,se
## 14: response
## 15: response
## 16: response,se
## 17: response,se
## 18: response,se
## 19: response
## 20: response
## 21: response
## predict_types
```

We can now pick the logistic one. **Note** that this essentially proxies our requests down to the `stats` package.

```
learner = mlr_learners$get("classif.log_reg")
```

Now we can finally solve the question, which is to simply use the model on all our data and return the accuracy metrics.

```
trainFullCred=learner$train(tskLogiFull)
print(learner$predict(tskLogiFull)$confusion)
```

```
##      truth
## response  No  Yes
##      No  9629  225
##      Yes   38  108
```

```
measure = msr("classif.acc")
print(learner$predict(tskLogiFull)$score(measure))
```

```
## classif.acc
##      0.9737
```

Note that this style of working with objects does not really utilize the familiar `%>%` interface.

The `caret` package still has neater default metrics so we will use that as well.

```
confusionMatrix(learner$predict(tskLogiFull)$response,defDat$default)
```

```
## Confusion Matrix and Statistics
##
##           Reference
## Prediction   No   Yes
##           No  9629  225
##           Yes   38  108
##
##           Accuracy : 0.9737
##           95% CI : (0.9704, 0.9767)
##           No Information Rate : 0.9667
##           P-Value [Acc > NIR] : 3.067e-05
##
##           Kappa : 0.4396
##
## McNemar's Test P-Value : < 2.2e-16
##
##           Sensitivity : 0.9961
##           Specificity : 0.3243
##           Pos Pred Value : 0.9772
##           Neg Pred Value : 0.7397
##           Prevalence : 0.9667
##           Detection Rate : 0.9629
##           Detection Prevalence : 0.9854
##           Balanced Accuracy : 0.6602
##
##           'Positive' Class : No
##
```

```
autoplot(learner$predict(tskLogiFull))
```

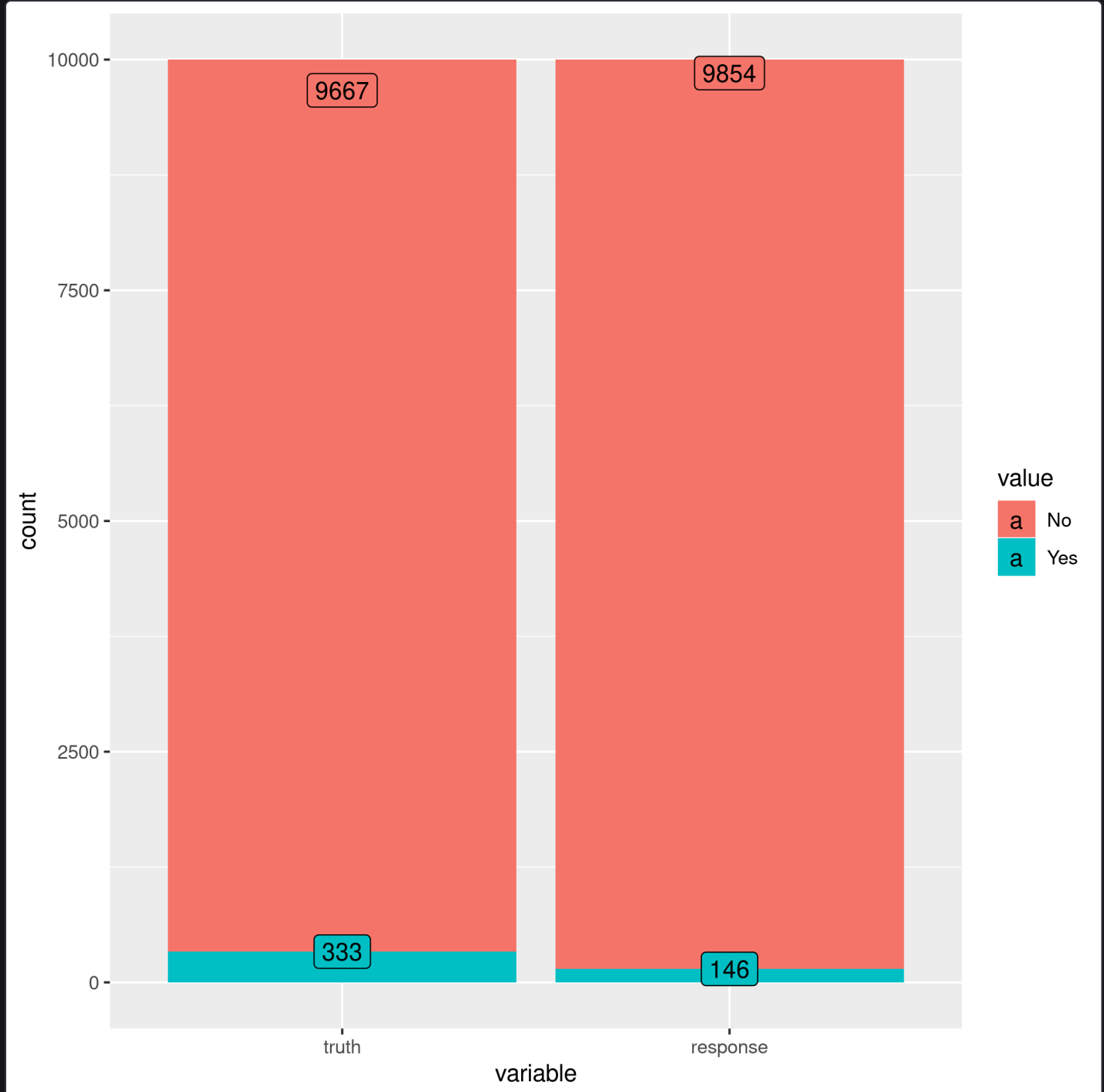


Figure 3: Autoplot results

We can get some other plots as well, but we need our probabilities to be returned.

```
# For ROC curves
lrnprob = lrn("classif.log_reg",predict_type="prob")
lrnprob$train(tskLogiFull)
autoplot(lrnprob$predict(tskLogiFull),type="roc")
```


ROC - P: 333, N: 9667

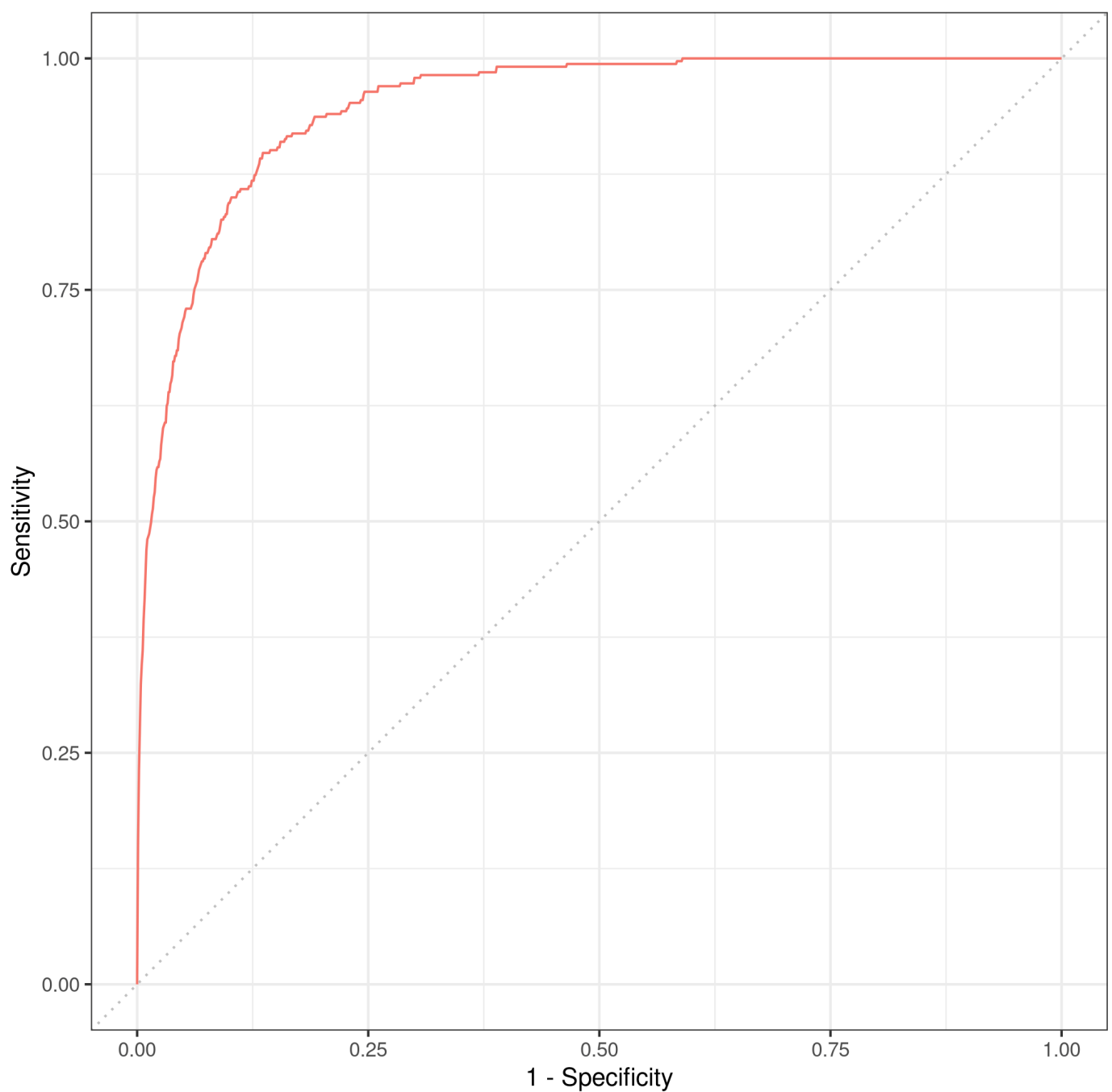


Figure 4: ROC curve

b) Validation Sets with mlr3

Though the question seems to require a manual validation set generation and thresholding, we can simply use the defaults.

```
train_set = sample(tskLogiFull$nrow, 0.8 * tskLogiFull$nrow)
test_set = setdiff(seq_len(tskLogiFull$nrow), train_set)
learner$train(tskLogiFull, row_ids=train_set)
confusionMatrix(learner$predict(tskLogiFull, row_ids=test_set)$response, defDat[,-train_set,]$default)
```

```
## Confusion Matrix and Statistics
##
##           Reference
## Prediction  No  Yes
##           No 1921  47
##           Yes   9  23
##
##           Accuracy : 0.972
##           95% CI : (0.9638, 0.9788)
##           No Information Rate : 0.965
##           P-Value [Acc > NIR] : 0.04663
##
##           Kappa : 0.4387
##
## McNemar's Test P-Value : 7.641e-07
##
##           Sensitivity : 0.9953
##           Specificity : 0.3286
##           Pos Pred Value : 0.9761
##           Neg Pred Value : 0.7188
##           Prevalence : 0.9650
##           Detection Rate : 0.9605
##           Detection Prevalence : 0.9840
##           Balanced Accuracy : 0.6620
##
##           'Positive' Class : No
##
```

For a reasonable comparison, we will demonstrate a standard approach as well. In this instance we will not use `caret` to ensure that our class distribution in the train and test sets are not sampled to remain the same.

```
trainNoCaret<-sample(nrow(defDat), size = floor(.8*nrow(defDat)), replace = F)
glm.fit=glm(default~income+balance,data=defDat,family=binomial,subset=trainNoCaret)
glm.probs<-predict(glm.fit,defDat[-trainNoCaret,],type="response")
glm.preds<-ifelse(glm.probs < 0.5, "No", "Yes")
confusionMatrix(glm.preds %>% factor,defDat[-trainNoCaret,]$default)
```

```
## Confusion Matrix and Statistics
##
##           Reference
## Prediction  No  Yes
##           No 1930  46
##           Yes   6  18
##
##           Accuracy : 0.974
##           95% CI : (0.966, 0.9805)
##           No Information Rate : 0.968
##           P-Value [Acc > NIR] : 0.06859
##
##           Kappa : 0.3986
##
## McNemar's Test P-Value : 6.362e-08
##
##           Sensitivity : 0.9969
##           Specificity : 0.2812
##           Pos Pred Value : 0.9767
##           Neg Pred Value : 0.7500
##           Prevalence : 0.9680
##           Detection Rate : 0.9650
##           Detection Prevalence : 0.9880
##           Balanced Accuracy : 0.6391
##
##           'Positive' Class : No
##
```

Since the two approaches use different samples there is a little variation, but we can see that the accuracy is essentially the same.

c) 3-fold cross validation

As per the question, we can repeat the block above three times, or extract it into a function which takes a seed value and run that three times. Either way, here we will present the `mlr3` approach to cross validation and resampling.

```
rr = resample(tskLogiFull, lrnprob, rsmp("cv", folds = 3))
```

```
## INFO [22:12:30.025] Applying learner 'classif.log_reg' on task 'credit' (iter 1/3)
## INFO [22:12:30.212] Applying learner 'classif.log_reg' on task 'credit' (iter 2/3)
## INFO [22:12:30.360] Applying learner 'classif.log_reg' on task 'credit' (iter 3/3)
```

```
autoplot(rr,type="roc")
```

ROC

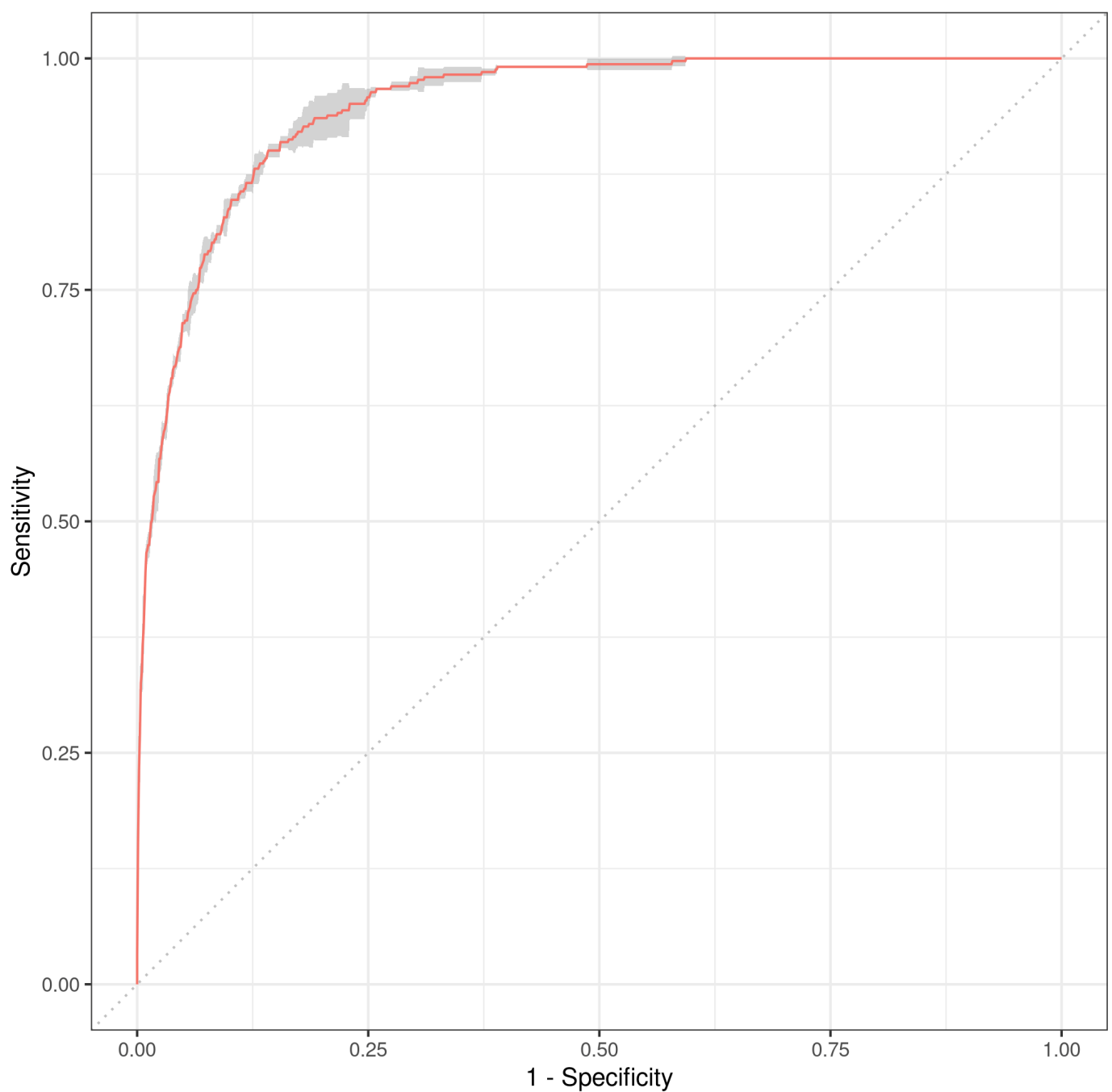


Figure 5: Resampled ROC curve

We might want the average as well.

```
rr$aggregate(msr("classif.ce")) %>% print
```

```
## classif.ce  
## 0.02630035
```

Adding Student as a dummy variable

We will stick to the `mlr3` approach because it is faster.

```
redDat2<-defDat %>% mutate(student=as.numeric(defDat$student))  
tskLogi2=TaskClassif$new(id="credit",backend=redDat2,target="default")  
print(tskLogi2)
```

```
## <TaskClassif:credit> (10000 x 4)
## * Target: default
## * Properties: twoclass
## * Features (3):
##   - dbl (3): balance, income, student
```

```
autoplot(tskLogi2,type="pairs")
```

```
## `stat_bin()` using `bins = 30`. Pick better value with `binwidth`.
## `stat_bin()` using `bins = 30`. Pick better value with `binwidth`.
## `stat_bin()` using `bins = 30`. Pick better value with `binwidth`.
```

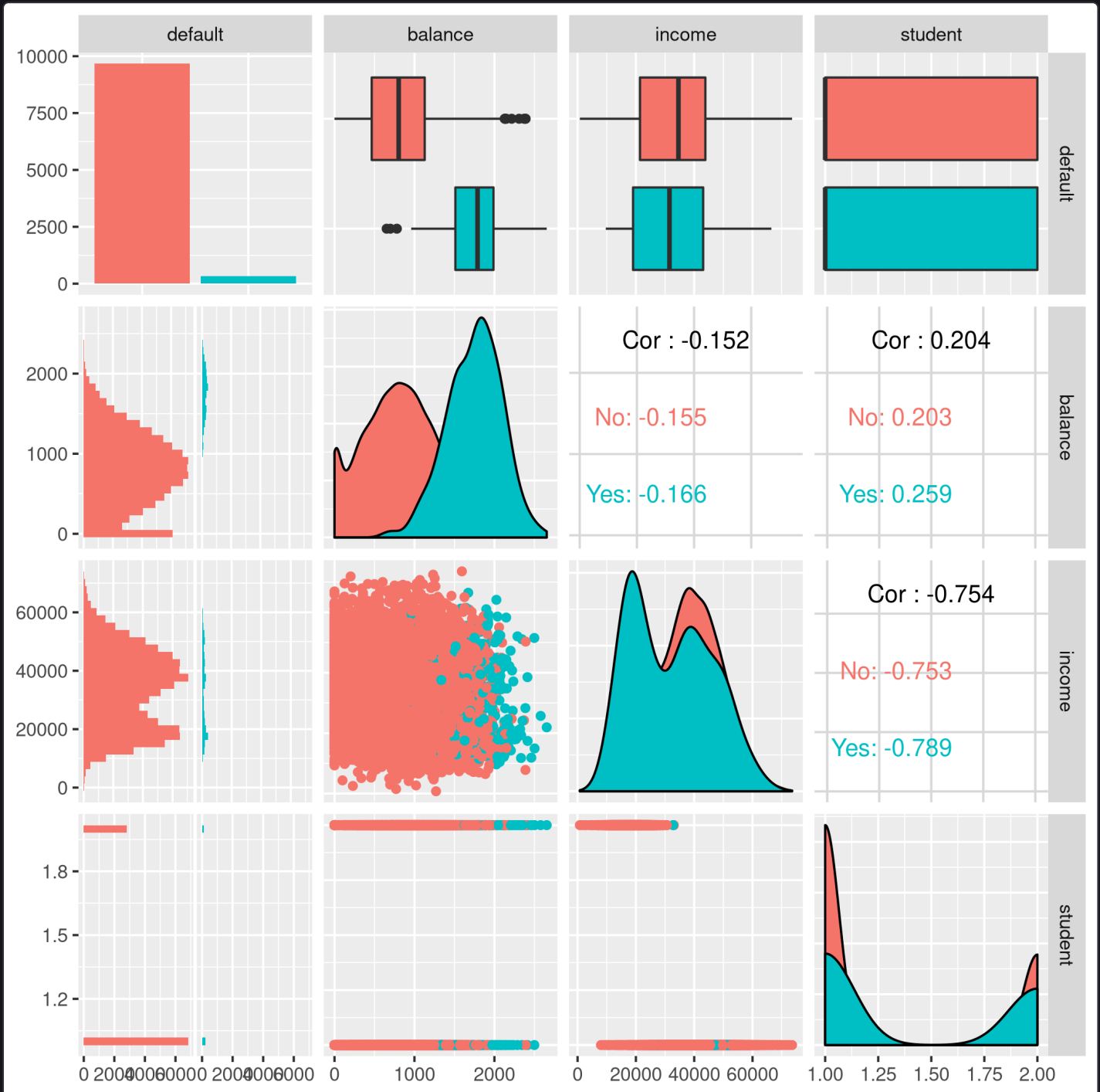


Figure 6: Logistic regression pairs data

This gives us a visual indicator and premonition that we might not be getting incredible results with our new variable in the mix, but we should still work it through.

```
confusionMatrix(lrnprob$predict(tskLogi2)$response,defDat$default)
```

```
## Confusion Matrix and Statistics
##
##           Reference
## Prediction  No  Yes
##           No 9629 225
##           Yes  38 108
##
##           Accuracy : 0.9737
##           95% CI : (0.9704, 0.9767)
##           No Information Rate : 0.9667
##           P-Value [Acc > NIR] : 3.067e-05
##
##           Kappa : 0.4396
##
## McNemar's Test P-Value : < 2.2e-16
##
##           Sensitivity : 0.9961
##           Specificity : 0.3243
##           Pos Pred Value : 0.9772
##           Neg Pred Value : 0.7397
##           Prevalence : 0.9667
##           Detection Rate : 0.9629
##           Detection Prevalence : 0.9854
##           Balanced Accuracy : 0.6602
##
##           'Positive' Class : No
##
```

```
autoplot(lrnprob$predict(tskLogi2))
```

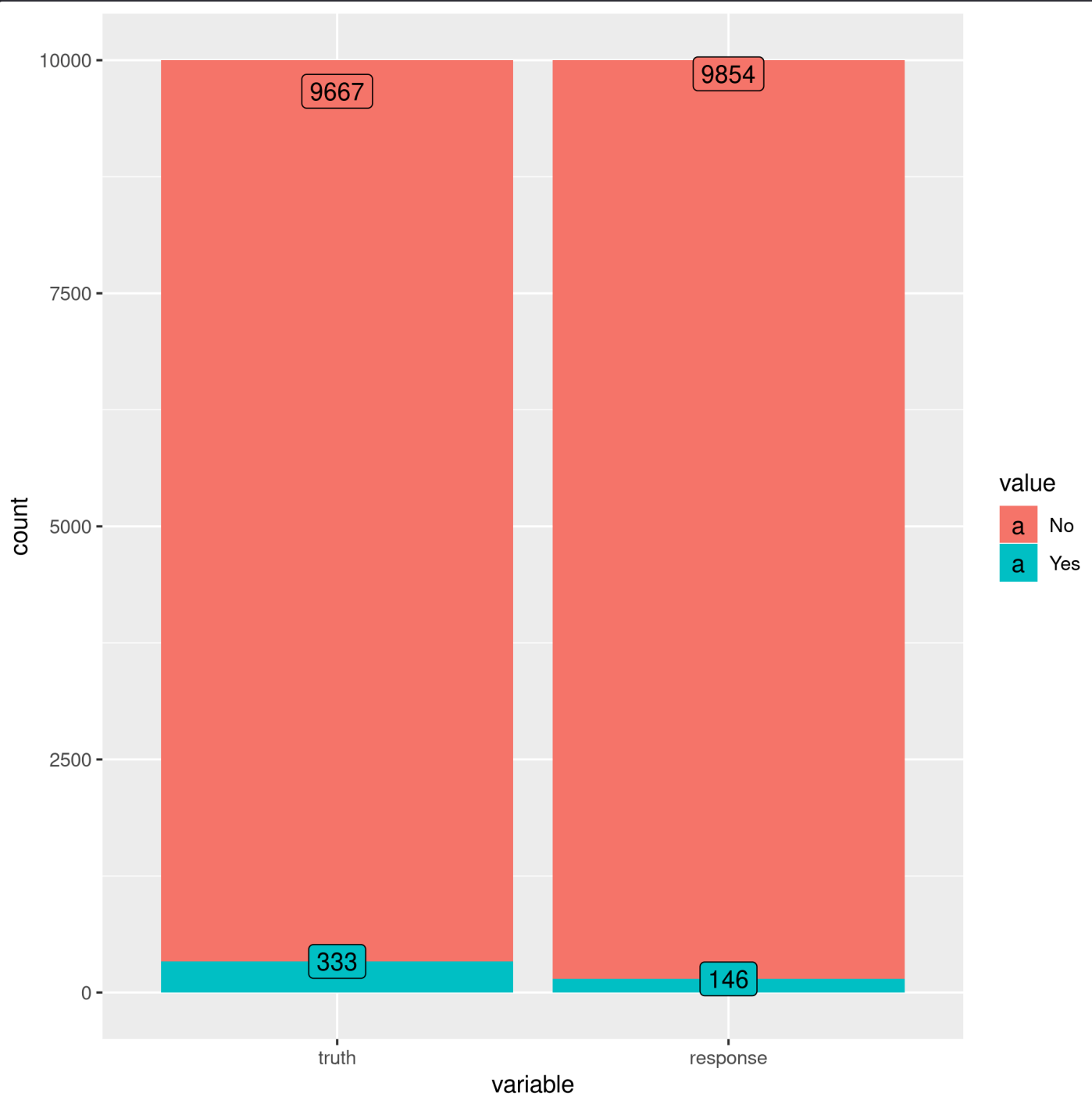


Figure 7: Autoplot figure

```
lnnprob$train(tskLogi2)
autoplot(lnnprob$predict(tskLogi2), type="roc")
```

ROC - P: 333, N: 9667

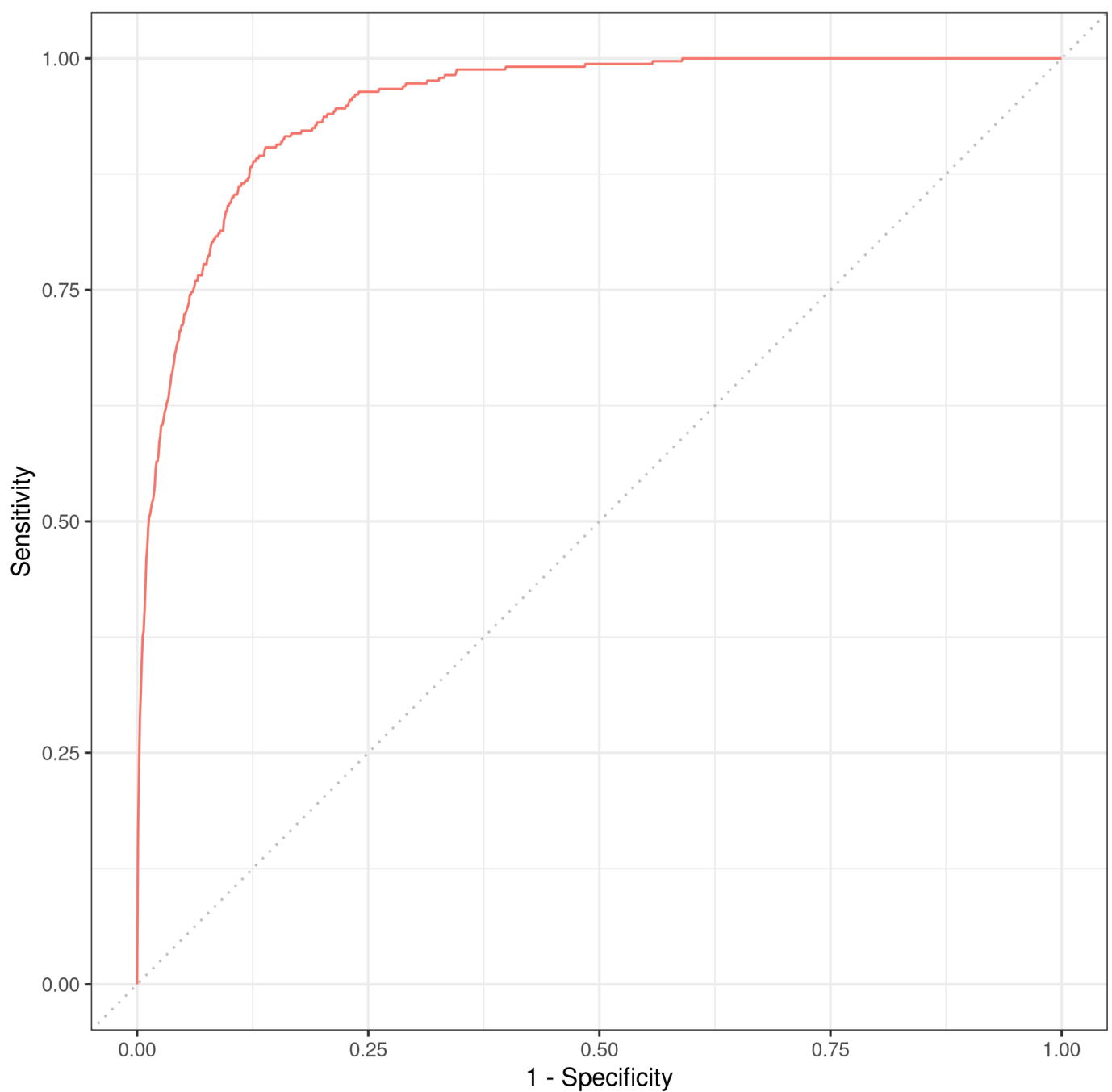


Figure 8: ROC plot

Although we have slightly better accuracy with the new variable, it needs to be compared to determine if it is worth further investigation.

With a three-fold validation approach,

```
library("gridExtra")
```

```
##  
## Attaching package: 'gridExtra'
```

```
## The following object is masked from 'package:dplyr':  
##  
## combine
```

```
rr2 = resample(tskLogi2, lrnprob, rsmp("cv", folds = 3))
```



```
## INFO [22:12:39.670] Applying learner 'classif.log_reg' on task 'credit' (iter 1/3)
## INFO [22:12:39.731] Applying learner 'classif.log_reg' on task 'credit' (iter 2/3)
## INFO [22:12:39.780] Applying learner 'classif.log_reg' on task 'credit' (iter 3/3)
```

```
wS<-autoplot(rr2)
nS<-autoplot(rr)
grid.arrange(wS,nS,ncol=2,bottom="With student (left) and without (right)")
```

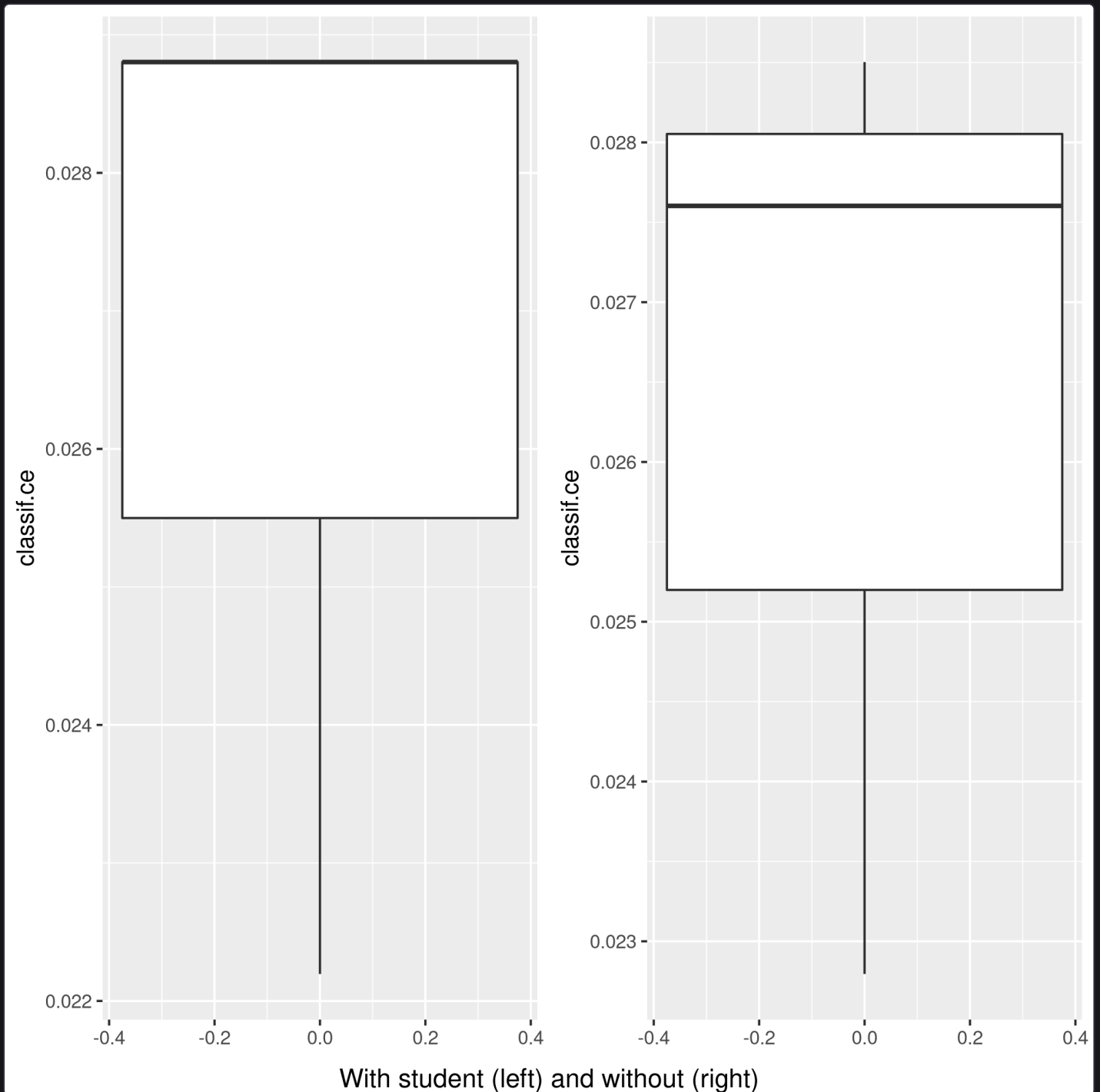


Figure 9: Plot of accuracy

Given the results, it is fair to say that adding the student data is useful in general.

Question 5.6 - Page 199

We continue to consider the use of a logistic regression model to predict the probability of `default` using `income` and `balance` on the `Default` data set. In particular, we will now compute estimates for the standard errors of the `income` and `balance` logistic regression coefficients in two different ways: (1) using the bootstrap, and (2) using the standard formula for computing the standard errors in the `glm()` function. Do not forget to set a random seed before beginning your analysis.

- (a)** Using the `summary()` and `glm()` functions, determine the estimated standard errors for the coefficients associated with `income` and `balance` in a multiple logistic regression model that uses both predictors.
- (b)** Write a function, `boot.fn()`, that takes as input the `Default` data set as well as an index of the observations, and that outputs the coefficient estimates for `income` and `balance` in the multiple logistic regression model.
- (c)** Use the `boot()` function together with your `boot.fn()` function to estimate the standard errors of the logistic regression coefficients for `income` and `balance`.
- (d)** Comment on the estimated standard errors obtained using the `glm()` function and using your bootstrap function.

Answer

This question is slightly more specific to the packages in the book so we will use them.

a) Fit summary

```
glm.fit %>% summary
```

```
##
## Call:
## glm(formula = default ~ income + balance, family = binomial,
##      data = defDat, subset = trainNoCaret)
##
## Deviance Residuals:
##      Min       1Q   Median       3Q      Max
## -2.1943  -0.1488  -0.0588  -0.0217   3.7058
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept) -1.150e+01  4.814e-01 -23.885  < 2e-16 ***
## income       2.288e-05  5.553e-06   4.121 3.78e-05 ***
## balance      5.593e-03  2.509e-04  22.295  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##      Null deviance: 2354.0  on 7999  degrees of freedom
## Residual deviance: 1283.6  on 7997  degrees of freedom
## AIC: 1289.6
##
## Number of Fisher Scoring iterations: 8
```

b) Function

```
boot.fn=function(data,subs){return(coef(glm(default~income+balance,data=data, family=binomial,subset=subs)))}
```

```
boot.fn(defDat,train_set) %>% print
```

```
##      (Intercept)      income      balance
## -1.136824e+01  1.846153e-05  5.576468e-03
```

```
glm(default~income+balance,data=defDat,family=binomial,subset=train_set) %>% summary
```

```
##
## Call:
## glm(formula = default ~ income + balance, family = binomial,
##      data = defDat, subset = train_set)
##
## Deviance Residuals:
##      Min       1Q   Median       3Q      Max
## -2.4280  -0.1465  -0.0582  -0.0218   3.7115
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept) -1.137e+01  4.813e-01 -23.618  < 2e-16 ***
## income      1.846e-05  5.553e-06   3.324 0.000886 ***
## balance     5.576e-03  2.529e-04  22.046  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##      Null deviance: 2313.6  on 7999  degrees of freedom
## Residual deviance: 1266.4  on 7997  degrees of freedom
## AIC: 1272.4
##
## Number of Fisher Scoring iterations: 8
```

We see that the statistics obtained from both are the same.

c) Bootstrap

The old fashioned way. `R` is the resample rate, `boot.fn` is the statistic used.

```
library(boot)
```

```
##
## Attaching package: 'boot'
```

```
## The following object is masked from 'package:lattice':
##
##      melanoma
```

```
boot(defDat,boot.fn,R=184) %>% print
```

```
##
## ORDINARY NONPARAMETRIC BOOTSTRAP
##
##
## Call:
## boot(data = defDat, statistic = boot.fn, R = 184)
##
##
## Bootstrap Statistics :
##      original      bias      std. error
## t1* -1.154047e+01 -1.407368e-02 4.073453e-01
## t2*  2.080898e-05 -6.386634e-08 4.720109e-06
## t3*  5.647103e-03  1.350950e-05 2.111547e-04
```

d) Comparison

- Clearly, there is not much difference in the standard error estimates

Var | Bootstrap | Summary |

| :-----: | ----- |

Intercept | 4.428026e-01 | 4.883e-01 |

income | 2.797011e-06 | 5.548e-06 |

balance | 2.423002e-04 | 2.591e-04 |

Question 5.8 - Page 200

We will now perform cross-validation on a simulated data set. **(a)** Generate a simulated data set as follows:

```
> set . seed (1)
> y = rnorm (100)
> x = rnorm (100)
> y =x -2\* x ^2+ rnorm (100)
```

In this data set, what is n and what is p ? Write out the model used to generate the data in equation form.

(b) Create a scatterplot of $\mathbf{X}$ against $\mathbf{Y}$. Comment on what you find.

(c) Set a random seed, and then compute the LOOCV errors that result from fitting the following four models using least squares:

1. $Y = \beta_0 + \beta_1 X + \eta$

2. $Y = \beta_0 + \beta_1 X + \beta_2 X^2 + \eta$

3. $Y = \beta_0 + \beta_1 X + \beta_2 X^2 + \beta_3 X^3 + \eta$

4. $Y = \beta_0 + \beta_1 X + \beta_2 X^2 + \beta_3 X^3 + \beta_4 X^4 + \eta$

Note you may find it helpful to use the `data.frame()` function to create a single data set containing both $\mathbf{X}$ and $\mathbf{Y}$.

(d) Repeat (c) using another random seed, and report your results. Are your results the same as what you got in (c)? Why?

(e) Which of the models in (c) had the smallest LOOCV error? Is this what you expected? Explain your answer.

(f) Comment on the statistical significance of the coefficient estimates that results from fitting each of the models in (c) using least squares. Do these results agree with the conclusions drawn based on the cross-validation results?

Answer

a) Modeling data

```
set.seed(1)
y <- rnorm(100)
x <- rnorm(100)
y <- x - 2*x^2 + rnorm(100)
```

Clearly:

- Our equation is $y = x - 2x^2 + \epsilon$ where *epsilon* is normally distributed from 100 samples
- We have $n = 100$ observations
- $p = 2$ where p is the number of features

b) Visual inspection

```
qplot(x,y)
```

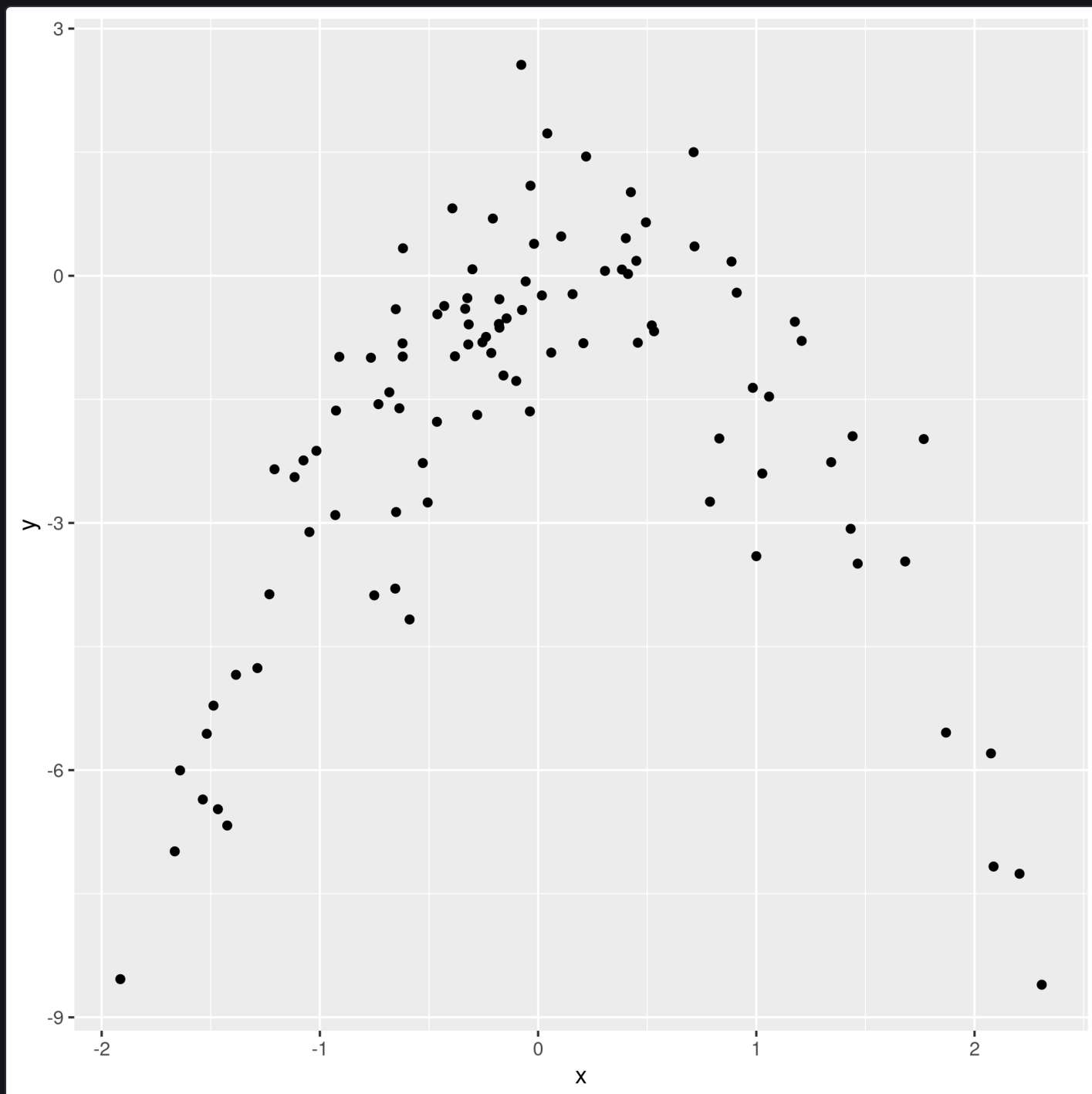


Figure 10: Model data plot

We observe that the data is quadratic, as we also know from the generating function, which was a quadratic equation plus normally distributed noise.

c) Least squares fits

Not very important, but here we use the `caret` form.

```
pow=function(x,y){return(x^y)}  
dfDat <- data.frame(y,x,x2=pow(x,2),x3=pow(x,3),x4=pow(x,4))
```

We might have also just used `poly(x,n)` to skip making the data frame.

We will set our resampling method as follows:

```
fitControl<-trainControl(method="LOOCV")
```

```
train(y~x,data=dfDat,trControl=fitControl,method="lm") %>% print
```

```
## Linear Regression  
##  
## 100 samples  
## 1 predictor  
##  
## No pre-processing  
## Resampling: Leave-One-Out Cross-Validation  
## Summary of sample sizes: 99, 99, 99, 99, 99, 99, ...  
## Resampling results:  
##  
## RMSE      Rsquared    MAE  
## 2.427134  0.05389864  1.878566  
##  
## Tuning parameter 'intercept' was held constant at a value of TRUE
```

```
train(y~x+x2,data=dfDat,trControl=fitControl,method="lm") %>% print
```

```
## Linear Regression  
##  
## 100 samples  
## 2 predictor  
##  
## No pre-processing  
## Resampling: Leave-One-Out Cross-Validation  
## Summary of sample sizes: 99, 99, 99, 99, 99, 99, ...  
## Resampling results:  
##  
## RMSE      Rsquared    MAE  
## 1.042399  0.8032414  0.8029942  
##  
## Tuning parameter 'intercept' was held constant at a value of TRUE
```

```
train(y~x+x2+x3,data=dfDat,trControl=fitControl,method="lm") %>% print
```

```
## Linear Regression
##
## 100 samples
##    3 predictor
##
## No pre-processing
## Resampling: Leave-One-Out Cross-Validation
## Summary of sample sizes: 99, 99, 99, 99, 99, 99, ...
## Resampling results:
##
##    RMSE      Rsquared   MAE
##    1.050041   0.8003517   0.8073024
##
## Tuning parameter 'intercept' was held constant at a value of TRUE
```

```
train(y~x+x2+x3+x4,data=dfDat,trControl=fitControl,method="lm") %>% print
```

```
## Linear Regression
##
## 100 samples
##    4 predictor
##
## No pre-processing
## Resampling: Leave-One-Out Cross-Validation
## Summary of sample sizes: 99, 99, 99, 99, 99, 99, ...
## Resampling results:
##
##    RMSE      Rsquared   MAE
##    1.055828   0.7982111   0.8150296
##
## Tuning parameter 'intercept' was held constant at a value of TRUE
```

d) Seeding effects

```
set.seed(1995)
```

```
train(y~x,data=dfDat,trControl=fitControl,method="lm") %>% print
```

```
## Linear Regression
##
## 100 samples
##    1 predictor
##
## No pre-processing
## Resampling: Leave-One-Out Cross-Validation
## Summary of sample sizes: 99, 99, 99, 99, 99, 99, ...
## Resampling results:
##
##    RMSE      Rsquared   MAE
##    2.427134   0.05389864   1.878566
##
## Tuning parameter 'intercept' was held constant at a value of TRUE
```

```
train(y~x+x2,data=dfDat,trControl=fitControl,method="lm") %>% print
```

```
## Linear Regression
##
## 100 samples
## 2 predictor
##
## No pre-processing
## Resampling: Leave-One-Out Cross-Validation
## Summary of sample sizes: 99, 99, 99, 99, 99, 99, ...
## Resampling results:
##
## RMSE      Rsquared  MAE
## 1.042399  0.8032414  0.8029942
##
## Tuning parameter 'intercept' was held constant at a value of TRUE
```

```
train(y~x+x2+x3,data=dfDat,trControl=fitControl,method="lm") %>% print
```

```
## Linear Regression
##
## 100 samples
## 3 predictor
##
## No pre-processing
## Resampling: Leave-One-Out Cross-Validation
## Summary of sample sizes: 99, 99, 99, 99, 99, 99, ...
## Resampling results:
##
## RMSE      Rsquared  MAE
## 1.050041  0.8003517  0.8073024
##
## Tuning parameter 'intercept' was held constant at a value of TRUE
```

```
train(y~x+x2+x3+x4,data=dfDat,trControl=fitControl,method="lm") %>% print
```

```
## Linear Regression
##
## 100 samples
## 4 predictor
##
## No pre-processing
## Resampling: Leave-One-Out Cross-Validation
## Summary of sample sizes: 99, 99, 99, 99, 99, 99, ...
## Resampling results:
##
## RMSE      Rsquared  MAE
## 1.055828  0.7982111  0.8150296
##
## Tuning parameter 'intercept' was held constant at a value of TRUE
```

We note that there is no change on varying the seed because LOOCV is exhaustive and uses n folds for each observation.

e) Analysis

```
train(y~x,data=dfDat %>% subset(select=c(y,x)),trControl=fitControl,method="lm") %>% print
```



```
## Linear Regression
##
## 100 samples
## 1 predictor
##
## No pre-processing
## Resampling: Leave-One-Out Cross-Validation
## Summary of sample sizes: 99, 99, 99, 99, 99, 99, ...
## Resampling results:
##
## RMSE      Rsquared    MAE
## 2.427134  0.05389864  1.878566
##
## Tuning parameter 'intercept' was held constant at a value of TRUE
```

```
train(y~poly(x,2),data=dfDat %>% subset(select=c(y,x)),trControl=fitControl,method="lm") %>% print
```

```
## Linear Regression
##
## 100 samples
## 1 predictor
##
## No pre-processing
## Resampling: Leave-One-Out Cross-Validation
## Summary of sample sizes: 99, 99, 99, 99, 99, 99, ...
## Resampling results:
##
## RMSE      Rsquared    MAE
## 1.042399  0.8032414  0.8029942
##
## Tuning parameter 'intercept' was held constant at a value of TRUE
```

```
train(y~poly(x,3),data=dfDat %>% subset(select=c(y,x)),trControl=fitControl,method="lm") %>% print
```

```
## Linear Regression
##
## 100 samples
## 1 predictor
##
## No pre-processing
## Resampling: Leave-One-Out Cross-Validation
## Summary of sample sizes: 99, 99, 99, 99, 99, 99, ...
## Resampling results:
##
## RMSE      Rsquared    MAE
## 1.050041  0.8003517  0.8073024
##
## Tuning parameter 'intercept' was held constant at a value of TRUE
```

```
train(y~poly(x,4),data=dfDat %>% subset(select=c(y,x)),trControl=fitControl,method="lm") %>% print
```

```
## Linear Regression
##
## 100 samples
## 1 predictor
##
## No pre-processing
## Resampling: Leave-One-Out Cross-Validation
## Summary of sample sizes: 99, 99, 99, 99, 99, 99, ...
## Resampling results:
##
## RMSE      Rsquared   MAE
## 1.055828  0.7982111  0.8150296
##
## Tuning parameter 'intercept' was held constant at a value of TRUE
```

Clearly the quadratic polynomial has the lowest error, which makes sense given how the data was generated.

f) Statistical significance

```
train(y~x,data=dfDat %>% subset(select=c(y,x)),trControl=fitControl,method="lm") %>% summary %>% print
```

```
##
## Call:
## lm(formula = .outcome ~ ., data = dat)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -7.3469 -0.9275  0.8028  1.5608  4.3974
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  -1.8185     0.2364  -7.692 1.14e-11 ***
## x              0.2430     0.2479   0.981  0.329
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 2.362 on 98 degrees of freedom
## Multiple R-squared:  0.009717, Adjusted R-squared:  -0.0003881
## F-statistic: 0.9616 on 1 and 98 DF, p-value: 0.3292
```

```
train(y~poly(x,2),data=dfDat %>% subset(select=c(y,x)),trControl=fitControl,method="lm") %>% summary %>% print
```

```
##
## Call:
## lm(formula = .outcome ~ ., data = dat)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -2.89884 -0.53765  0.04135  0.61490  2.73607
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   -1.8277     0.1032  -17.704  <2e-16 ***
## `poly(x, 2)1`    2.3164     1.0324   2.244  0.0271 *
## `poly(x, 2)2`  -21.0586     1.0324 -20.399  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.032 on 97 degrees of freedom
## Multiple R-squared:  0.8128, Adjusted R-squared:  0.8089
## F-statistic: 210.6 on 2 and 97 DF,  p-value: < 2.2e-16
```

```
train(y~poly(x,3),data=dfDat %>% subset(select=c(y,x)),trControl=fitControl,method="lm") %>% summary %>% print
```

```
##
## Call:
## lm(formula = .outcome ~ ., data = dat)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -2.87250 -0.53881  0.02862  0.59383  2.74350
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   -1.8277     0.1037  -17.621  <2e-16 ***
## `poly(x, 3)1`    2.3164     1.0372   2.233  0.0279 *
## `poly(x, 3)2`  -21.0586     1.0372 -20.302  <2e-16 ***
## `poly(x, 3)3`   -0.3048     1.0372  -0.294  0.7695
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.037 on 96 degrees of freedom
## Multiple R-squared:  0.813, Adjusted R-squared:  0.8071
## F-statistic: 139.1 on 3 and 96 DF,  p-value: < 2.2e-16
```

```
train(y~poly(x,4),data=dfDat %>% subset(select=c(y,x)),trControl=fitControl,method="lm") %>% summary %>% print
```

```
##
## Call:
## lm(formula = .outcome ~ ., data = dat)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -2.8914 -0.5244  0.0749  0.5932  2.7796
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   -1.8277     0.1041  -17.549  <2e-16 ***
## `poly(x, 4)1`    2.3164     1.0415   2.224  0.0285 *
## `poly(x, 4)2`  -21.0586     1.0415 -20.220  <2e-16 ***
## `poly(x, 4)3`   -0.3048     1.0415  -0.293  0.7704
## `poly(x, 4)4`   -0.4926     1.0415  -0.473  0.6373
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.041 on 95 degrees of freedom
## Multiple R-squared:  0.8134, Adjusted R-squared:  0.8055
## F-statistic: 103.5 on 4 and 95 DF,  p-value: < 2.2e-16
```

- Clearly, the second order terms are the most significant, as expected

Question 5.9 - Page 201

We will now consider the Boston housing data set, from the MASS library.

- (a)** Based on this data set, provide an estimate for the population mean of `medv`. Call this estimate $\hat{\mu}$.
- (b)** Provide an estimate of the standard error of $\hat{\mu}$. Interpret this result. *Hint: We can compute the standard error of the sample mean by dividing the sample standard deviation by the square root of the number of observations.*
- (c)** Now estimate the standard error of $\hat{\mu}$ using the bootstrap. How does this compare to your answer from (b)?
- (d)** Based on your bootstrap estimate from (c), provide a 95 % confidence interval for the mean of `medv`. Compare it to the results obtained using `t.test(Boston$medv)`. *Hint: You can approximate a 95 % confidence interval using the formula $[\hat{\mu} - 2SE(\hat{\mu}), \hat{\mu} + 2SE(\hat{\mu})]$.*
- (e)** Based on this data set, provide an estimate, $\mu_{med}^{\wedge}$, for the median value of `medv` in the population.
- (f)** We now would like to estimate the standard error of $\hat{\mu}_{med}$. Unfortunately, there is no simple formula for computing the standard error of the median. Instead, estimate the standard error of the median using the bootstrap. Comment on your findings.
- (g)** Based on this data set, provide an estimate for the tenth percentile of `medv` in Boston suburbs. Call this quantity $\mu_{0.1}^{\wedge}$. (You can use the `quantile()` function.)
- (h)** Use the bootstrap to estimate the standard error of $\mu_{0.1}^{\wedge}$. Comment on your findings.

Answer

```
boston<-MASS::Boston
```

- Reminder

```
boston %>% summary %>% print
```

```
##           crim           zn           indus           chas
## Min.      : 0.00632   Min.      : 0.00   Min.      : 0.46   Min.      :0.00000
## 1st Qu.: 0.08204   1st Qu.: 0.00   1st Qu.: 5.19   1st Qu.:0.00000
## Median : 0.25651   Median : 0.00   Median : 9.69   Median :0.00000
## Mean      : 3.61352   Mean      :11.36   Mean      :11.14   Mean      :0.06917
## 3rd Qu.: 3.67708   3rd Qu.:12.50   3rd Qu.:18.10   3rd Qu.:0.00000
## Max.      :88.97620   Max.      :100.00   Max.      :27.74   Max.      :1.00000
##           nox           rm           age           dis
## Min.      :0.3850   Min.      :3.561   Min.      : 2.90   Min.      : 1.130
## 1st Qu.:0.4490   1st Qu.:5.886   1st Qu.: 45.02   1st Qu.: 2.100
## Median :0.5380   Median :6.208   Median : 77.50   Median : 3.207
## Mean      :0.5547   Mean      :6.285   Mean      : 68.57   Mean      : 3.795
## 3rd Qu.:0.6240   3rd Qu.:6.623   3rd Qu.: 94.08   3rd Qu.: 5.188
## Max.      :0.8710   Max.      :8.780   Max.      :100.00   Max.      :12.127
##           rad           tax           ptratio         black
## Min.      : 1.000   Min.      :187.0   Min.      :12.60   Min.      : 0.32
## 1st Qu.: 4.000   1st Qu.:279.0   1st Qu.:17.40   1st Qu.:375.38
## Median : 5.000   Median :330.0   Median :19.05   Median :391.44
## Mean      : 9.549   Mean      :408.2   Mean      :18.46   Mean      :356.67
## 3rd Qu.:24.000   3rd Qu.:666.0   3rd Qu.:20.20   3rd Qu.:396.23
## Max.      :24.000   Max.      :711.0   Max.      :22.00   Max.      :396.90
##           lstat           medv
## Min.      : 1.73   Min.      : 5.00
## 1st Qu.: 6.95   1st Qu.:17.02
## Median :11.36   Median :21.20
## Mean      :12.65   Mean      :22.53
## 3rd Qu.:16.95   3rd Qu.:25.00
## Max.      :37.97   Max.      :50.00
```

```
boston %>% str %>% print
```

```
## 'data.frame':   506 obs. of  14 variables:
## $ crim      : num  0.00632 0.02731 0.02729 0.03237 0.06905 ...
## $ zn        : num  18 0 0 0 0 12.5 12.5 12.5 12.5 ...
## $ indus     : num  2.31 7.07 7.07 2.18 2.18 2.18 7.87 7.87 7.87 7.87 ...
## $ chas      : int   0 0 0 0 0 0 0 0 0 0 ...
## $ nox       : num  0.538 0.469 0.469 0.458 0.458 0.458 0.524 0.524 0.524 0.524 ...
## $ rm        : num  6.58 6.42 7.18 7 7.15 ...
## $ age       : num  65.2 78.9 61.1 45.8 54.2 58.7 66.6 96.1 100 85.9 ...
## $ dis       : num  4.09 4.97 4.97 6.06 6.06 ...
## $ rad       : int   1 2 2 3 3 3 5 5 5 5 ...
## $ tax       : num  296 242 242 222 222 222 311 311 311 311 ...
## $ ptratio   : num  15.3 17.8 17.8 18.7 18.7 18.7 15.2 15.2 15.2 15.2 ...
## $ black     : num  397 397 393 395 397 ...
## $ lstat     : num  4.98 9.14 4.03 2.94 5.33 ...
## $ medv      : num  24 21.6 34.7 33.4 36.2 28.7 22.9 27.1 16.5 18.9 ...
## NULL
```

a) Mean

```
muhat=boston$medv %>% mean()
print(muhat)
```

```
## [1] 22.53281
```

b) Standard error

Recall that $SE = \frac{SD}{\sqrt{N_{obs}}}$

```
boston$medv %>% sd/(nrow(boston)^0.5) %>% print
```

```
## [1] 22.49444
```

```
## [1] 0.4088611
```

c) Bootstrap estimate

```
library(boot)
myMean<-function(frame,ind){return(mean(frame[ind]))}
```

```
boot(boston$medv,myMean,R=184) %>% print
```

```
##
## ORDINARY NONPARAMETRIC BOOTSTRAP
##
##
## Call:
## boot(data = boston$medv, statistic = myMean, R = 184)
##
##
## Bootstrap Statistics :
##      original      bias    std. error
## t1* 22.53281 0.03451839    0.409621
```

We see that the bootstrapped error over 184 samples is 0.4341499 while without it we had 0.4088611 which is similar enough.

d) Confidence intervals with bootstrap and t.test

```
boston$medv %>% t.test %>% print
```

```
##
## One Sample t-test
##
## data: .
## t = 55.111, df = 505, p-value < 2.2e-16
## alternative hypothesis: true mean is not equal to 0
## 95 percent confidence interval:
## 21.72953 23.33608
## sample estimates:
## mean of x
## 22.53281
```

We can approximate this with what we already have

```
bRes=boot(boston$medv,myMean,R=184)
seBoot<-bRes$t %>% var %>% sqrt
xlow=muhat-2*(seBoot)
xhigh=muhat+2*(seBoot)
c(xlow,xhigh) %>% print
```

```
## [1] 21.72675 23.33887
```

Our intervals are also pretty close to each other.

e) Median

```
boston$medv %>% sort %>% median %>% print
```

```
## [1] 21.2
```

f) Median standard error

We can reuse the logic of the `myMean` function defined previously.

```
myMedian=function(data,ind){return(median(data[ind]))}
```

```
boston$medv %>% boot(myMedian,R=1500) %>% print
```

```
##  
## ORDINARY NONPARAMETRIC BOOTSTRAP  
##  
##  
## Call:  
## boot(data = ., statistic = myMedian, R = 1500)  
##  
##  
## Bootstrap Statistics :  
##      original      bias    std. error  
## t1*      21.2 -0.03773333    0.387315
```

We see that the standard error is `0.3767072`.

g) Tenth percentile

```
mu0one<-boston$medv %>% quantile(c(0.1))  
print(mu0one)
```

```
##      10%  
## 12.75
```

h) Bootstrap

Once again.

```
myQuant=function(data,ind){return(quantile(data[ind],0.1))}
```

```
boston$medv %>% boot(myQuant,R=500) %>% print
```

```
##  
## ORDINARY NONPARAMETRIC BOOTSTRAP  
##  
##  
## Call:  
## boot(data = ., statistic = myQuant, R = 500)  
##  
##  
## Bootstrap Statistics :  
##      original  bias    std. error  
## t1*      12.75 -0.0095   0.4951415
```

The standard error is 0.5024526

Introduction to Statistical Learning: with Applications in R.
Berlin, Germany: Springer Science & Business Media. [^fn:2]: Lang et al., (2019). mlr3: A modern object-oriented machine
learning framework in R. Journal of Open Source Software, 4(44),
1903, <<https://doi.org/10.21105/joss.01903>>